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\\Research

Investigators - validated on FIT

Frederick

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\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\localizer

TA: 15 sec Coil Selection: Auto Voxel Size: 1.1×1.1×10.0 mm³ Acc:: None Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
TR	18.0 ms
TE	3.80 ms
Averages	2
Concatenations	3
AutoAlign	---

Contrast - Common

TR	18.0 ms
TE	3.80 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	40 deg

Contrast - Common

Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
Base Resolution	256
Phase Resolution	50 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Allowed

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	1
Distance Factor	20 %
Position	Isocenter

Geometry - Common

Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
TR	18.0 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	3

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.0 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	280 mm
FOV Phase	100.0 %
Phase Resolution	50 %

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1

Inline - Subtraction

StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off
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Inline - MapIt

MapIt	None
Flip Angle	40 deg
Measurements	1
Contrasts	1
TE	3.80 ms
TR	18.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	fl_rr
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Flow Compensation	Read
Bandwidth	320 Hz/Px
Asymmetric Echo	Allowed
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\vessel_scout_head

TA: 22 sec Coil Selection: Auto Voxel Size: 1.2×1.2×5.0 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	3rd Segment
Inline Movie	Off

Resolution - Common

Base Resolution	256
Phase Resolution	70 %
Slice Resolution	50 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm
TR	32.3 ms
TE	5.89 ms
Averages	1
Concatenations	1
AutoAlign	---

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
Distortion Correction	2D
Normalize	Prescan
Image Filter	On

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	24
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm
TR	32.3 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	1

Contrast - Common

TR	32.3 ms
TE	5.89 ms
Flip Angle	10 deg
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	300 mm
FOV Phase	68.8 %
Slice Thickness	5.00 mm

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm

Geometry - AutoAlign

Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Special Saturation	None
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
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System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	32.3 ms
Segments	1

Physio - Signal

Concatenations	1
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Flow

Flow Mode	Free
Encodings	2
Velocity Enc. 1	30 cm/s
Velocity Enc. 2	30 cm/s
Direction 1	F >> H
Direction 2	Through Plane
Rephased Images	On
Magnitude Images	Off
Magnitude Sum	On
Phase Images	Off

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	pc
Dimension	3D
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	349 Hz/Px
Asymmetric Echo	Weak
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Assistant

SAR Assistant	Off
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\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\tof_fl3d_tra_p2

TA: 8:19 min Coil Selection: Auto Voxel Size: 0.4x0.4x0.5 mm³ Acc:: 2.0 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	7
Distance Factor	-20 %
Position	R0.8 A10.6 H0.4 mm
Orientation	T > C1.9 > S0.1
Phase Encoding Dir.	R >> L
Slices per Slab	60
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	200 mm
FOV Phase	90.6 %
Slice Thickness	0.50 mm
TR	17.54 ms
TE	2.90 ms
Averages	1
Concatenations	7
AutoAlign	---

Contrast - Common

TR	17.54 ms
TE	2.90 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	18 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Each Measurement
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	200 mm
FOV Phase	90.6 %
Slice Thickness	0.50 mm
Base Resolution	256
Phase Resolution	95 %
Slice Resolution	50 %
Trajectory	Cartesian
Interpolation	2.00

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	2.0
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	7
Distance Factor	-20 %
Position	R0.8 A10.6 H0.4 mm
Orientation	T > C1.9 > S0.1
Phase Encoding Dir.	R >> L
Slices per Slab	60
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	200 mm
FOV Phase	90.6 %
Slice Thickness	0.50 mm
TR	17.54 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	7

Geometry - AutoAlign

Slab Group	1
Position	R0.8 A10.6 H0.4 mm
Orientation	T > C1.9 > S0.1
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	R0.8 A10.6 H0.4
R	0.8 mm
A	10.6 mm
H	0.4 mm
Initial Orientation	T > C
T > C	1.90
> S	0.10
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
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System - Adjust Volume

Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	17.54 ms
Segments	1
Concatenations	7

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	90.6 %
Phase Resolution	95 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	7

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	2.90 ms
TR	17.54 ms

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	186 Hz/Px
Echo Spacing	8.19 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\tof_cs_acc12.0_large fast

TA: 3:51 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.4 mm³ Acc:: 12.0 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.40 mm
TR	20.31 ms
TE	4.30 ms
Averages	1
Concatenations	8
AutoAlign	Head > Brain

Contrast - Common

TR	20.31 ms
TE	4.30 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.40 mm
Base Resolution	272
Phase Resolution	100 %
Slice Resolution	64 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	12.0
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.40 mm
TR	20.31 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	8

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.31 ms
Segments	1
Concatenations	8

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	91.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	8

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	4.30 ms
TR	20.31 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	188 Hz/Px
Echo Spacing	9.65 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	TR
Max. TR	21.0 ms
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.5 mm³ Acc:: 10.3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	L0.0 A8.7 F8.7 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.50 mm
TR	18.22 ms
TE	3.13 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.22 ms
TE	3.13 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.50 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	L0.0 A8.7 F8.7 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.50 mm
TR	18.22 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A8.7 F8.7 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 A8.7 F8.7
L	0.0 mm
A	8.7 mm
F	8.7 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.22 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.13 ms
TR	18.22 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.52 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.5 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	L0.0 A8.7 F8.7 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.50 mm
TR	18.22 ms
TE	3.13 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.22 ms
TE	3.13 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.50 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	L0.0 A8.7 F8.7 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.50 mm
TR	18.22 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A8.7 F8.7 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 A8.7 F8.7
L	0.0 mm
A	8.7 mm
F	8.7 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.22 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.13 ms
TR	18.22 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.52 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.5 mm³ Acc:: 10.3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	L0.0 A8.7 F8.7 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.50 mm
TR	18.22 ms
TE	3.13 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.22 ms
TE	3.13 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.50 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	L0.0 A8.7 F8.7 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.50 mm
TR	18.22 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A8.7 F8.7 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 A8.7 F8.7
L	0.0 mm
A	8.7 mm
F	8.7 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.22 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.13 ms
TR	18.22 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.52 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.5 mm³ Acc:: 10.3 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	L0.0 A8.7 F8.7 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.50 mm
TR	18.22 ms
TE	3.13 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.22 ms
TE	3.13 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.50 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	L0.0 A8.7 F8.7 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.50 mm
TR	18.22 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A8.7 F8.7 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 A8.7 F8.7
L	0.0 mm
A	8.7 mm
F	8.7 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.22 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.13 ms
TR	18.22 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.52 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\tof_cs_acc10.3 fast

TA: 1:16 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.5 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	L0.0 A8.7 F8.7 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.50 mm
TR	18.22 ms
TE	3.13 ms
Averages	1
Concatenations	3
AutoAlign	Head > Brain

Contrast - Common

TR	18.22 ms
TE	3.13 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.50 mm
Base Resolution	304
Phase Resolution	99 %
Slice Resolution	67 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	3
Distance Factor	-20 %
Position	L0.0 A8.7 F8.7 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	220 mm
FOV Phase	89.5 %
Slice Thickness	0.50 mm
TR	18.22 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	3

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A8.7 F8.7 mm
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 A8.7 F8.7
L	0.0 mm
A	8.7 mm
F	8.7 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	18.22 ms
Segments	1
Concatenations	3

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	89.5 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	3

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.13 ms
TR	18.22 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	187 Hz/Px
Echo Spacing	8.52 ms
Asymmetric Echo	Weak
Optimization	Min. TE TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	Min. TE TR

\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\tof_fl3d_tra

TA: 5:33 min Coil Selection: Auto Voxel Size: 0.3×0.3×0.5 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	4
Distance Factor	-20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	200 mm
FOV Phase	90.6 %
Slice Thickness	0.50 mm
TR	21.00 ms
TE	3.42 ms
Averages	1
Concatenations	4
AutoAlign	Head > Brain

Contrast - Common

TR	21.00 ms
TE	3.42 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	200 mm
FOV Phase	90.6 %
Slice Thickness	0.50 mm
Base Resolution	384
Phase Resolution	95 %
Slice Resolution	50 %
Trajectory	Cartesian
Interpolation	2.00

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	7/8
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	4
Distance Factor	-20 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	200 mm
FOV Phase	90.6 %
Slice Thickness	0.50 mm
TR	21.00 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	4

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	21.00 ms
Segments	1
Concatenations	4

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	90.6 %
Phase Resolution	95 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	4

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.42 ms
TR	21.00 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	186 Hz/Px
Echo Spacing	9.32 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	16 deg
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\tof_fl3d_tra_large

TA: 5:12 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.5 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	6
Distance Factor	-18 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	0.50 mm
TR	19.00 ms
TE	3.50 ms
Averages	1
Concatenations	6
AutoAlign	Head > Brain

Contrast - Common

TR	19.00 ms
TE	3.50 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	15 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	0.50 mm
Base Resolution	256
Phase Resolution	94 %
Slice Resolution	50 %
Trajectory	Cartesian
Interpolation	2.00

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	32
Acceleration Factor 3D	1
Phase Partial Fourier	7/8
Slice Partial Fourier	7/8
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	6
Distance Factor	-18 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	40
Phase Oversampling	0 %
Slice Oversampling	20.0 %
FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	0.50 mm
TR	19.00 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	6

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	19.00 ms
Segments	1
Concatenations	6

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	180 mm
FOV Phase	100.0 %
Phase Resolution	94 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	6

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.50 ms
TR	19.00 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	199 Hz/Px
Echo Spacing	8.76 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\tof_f12d_sag_obl

TA: 2:52 min Coil Selection: Auto Voxel Size: 1.0×1.0×2.5 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	70
Distance Factor	-33 %
Position	Isocenter
Orientation	S > C10.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	2.5 mm
TR	17.30 ms
TE	6.15 ms
Averages	1
Concatenations	70
AutoAlign	---

Contrast - Common

TR	17.30 ms
TE	6.15 ms
TD	0.00 ms
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	60 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off
Reordering	Linear

Resolution - Common

FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	2.5 mm
Base Resolution	256
Phase Resolution	100 %
Trajectory	Cartesian
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	24
Phase Partial Fourier	Off
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	70
Distance Factor	-33 %
Position	Isocenter
Orientation	S > C10.0
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	100.0 %
Slice Thickness	2.5 mm
TR	17.30 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	70

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	S > C10.0
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
R	0.0 mm
P	0.0 mm
H	0.0 mm

Geometry - AutoAlign

Initial Orientation	S > C
S > C	10.00
> T	0.00
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Saturation Region	1
Thickness	80.00 mm
Position	L0.0 A12.7 F145.3 mm
Orientation	Transversal
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	17.30 ms
Segments	1
Concatenations	70

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	250 mm
FOV Phase	100.0 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	70

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	6.15 ms
TR	17.30 ms

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	2D
Sequence Type	Gre
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	219 Hz/Px
Echo Spacing	11.40 ms
Asymmetric Echo	Weak
Optimization	Min. TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	Min. TR

\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\tof_f12d_paracor

TA: 2:09 min Coil Selection: Auto Voxel Size: 1.0×1.0×2.5 mm³ Acc:: 2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	70
Distance Factor	-33 %
Position	Isocenter
Orientation	C > T-30.0
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	71.9 %
Slice Thickness	2.5 mm
TR	17.30 ms
TE	6.15 ms
Averages	1
Concatenations	70
AutoAlign	---

Contrast - Common

TR	17.30 ms
TE	6.15 ms
TD	0.00 ms
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	60 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Off
Reordering	Linear

Resolution - Common

FOV Read	250 mm
FOV Phase	71.9 %
Slice Thickness	2.5 mm
Base Resolution	256
Phase Resolution	100 %
Trajectory	Cartesian
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	2
Reference Lines PE	24
Phase Partial Fourier	Off
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	70
Distance Factor	-33 %
Position	Isocenter
Orientation	C > T-30.0
Phase Encoding Dir.	R >> L
Phase Oversampling	0 %
FOV Read	250 mm
FOV Phase	71.9 %
Slice Thickness	2.5 mm
TR	17.30 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	70

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	C > T-30.0
Phase Encoding Dir.	R >> L
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
F	0.0 mm

Geometry - AutoAlign

Initial Orientation	C > T
C > T	-30.00
> S	0.00
Initial Rotation	0.00 deg

Geometry - Navigator**Geometry - Saturation**

Saturation Region	1
Thickness	80.00 mm
Position	R6.4 A2.7 F107.1 mm
Orientation	T > C-0.1
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	17.30 ms
Segments	1
Concatenations	70

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	250 mm
FOV Phase	71.9 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	70

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	6.15 ms
TR	17.30 ms

Inline - MIP

MIP Sag	On
MIP Cor	On
MIP Tra	On
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	2D
Sequence Type	Gre
Excitation	Slice-sel.
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	219 Hz/Px
Echo Spacing	11.40 ms
Asymmetric Echo	Weak
Optimization	Min. TR
Define	Segments
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s
Optimization	Min. TR

\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\tof_cs_acc4.4

TA: 5:03 min Coil Selection: Auto Voxel Size: 0.4x0.4x0.4 mm³ Acc:: 4.4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms
TE	3.69 ms
Averages	1
Concatenations	4
AutoAlign	Head > Brain

Contrast - Common

TR	21.00 ms
TE	3.69 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	384
Phase Resolution	99 %
Slice Resolution	66 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	4.4
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	4

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F12.0
L	0.0 mm
P	0.0 mm
F	12.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	12 mm
Table Position	F
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	21.00 ms
Segments	1
Concatenations	4

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	90.6 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	4

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.69 ms
TR	21.00 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	186 Hz/Px
Echo Spacing	9.42 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\tof_cs_acc7.2

TA: 3:25 min Coil Selection: Auto Voxel Size: 0.4x0.4x0.4 mm³ Acc:: 7.2 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms
TE	3.69 ms
Averages	1
Concatenations	4
AutoAlign	Head > Brain

Contrast - Common

TR	21.00 ms
TE	3.69 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	384
Phase Resolution	99 %
Slice Resolution	66 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	7.2
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	4

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F12.0
L	0.0 mm
P	0.0 mm
F	12.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	12 mm
Table Position	F
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	21.00 ms
Segments	1
Concatenations	4

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	90.6 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	4

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.69 ms
TR	21.00 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	186 Hz/Px
Echo Spacing	9.42 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\tof_cs_acc10.3

TA: 2:40 min Coil Selection: Auto Voxel Size: 0.4×0.4×0.4 mm³ Acc:: 10.3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms
TE	3.69 ms
Averages	1
Concatenations	4
AutoAlign	Head > Brain

Contrast - Common

TR	21.00 ms
TE	3.69 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	70 %

Resolution - Common

FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
Base Resolution	384
Phase Resolution	99 %
Slice Resolution	66 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	10.3
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	4
Distance Factor	-21 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	23.1 %
FOV Read	220 mm
FOV Phase	90.6 %
Slice Thickness	0.40 mm
TR	21.00 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	4

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	L0.0 P0.0 F12.0
L	0.0 mm
P	0.0 mm
F	12.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	12 mm
Table Position	F
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	21.00 ms
Segments	1
Concatenations	4

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	220 mm
FOV Phase	90.6 %
Phase Resolution	99 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	4

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	3.69 ms
TR	21.00 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	186 Hz/Px
Echo Spacing	9.42 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	Flip Angle
Min Flip Angle	15 deg
Allowed Delay	0 s
Optimization	None

\\Research\\Investigators - validated on FIT\\Frederick\\Functional TOF\\tof_cs_acc12.0_large

TA: 3:51 min Coil Selection: Auto Voxel Size: 0.5×0.5×0.3 mm³ Acc:: 12.0 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms
TE	4.30 ms
Averages	1
Concatenations	8
AutoAlign	Head > Brain

Contrast - Common

TR	20.31 ms
TE	4.30 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle Mode	Constant
Flip Angle	20 deg
Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
Wrap-up Magn.	None
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1

Contrast - Dynamic

Multiple Series	Off
Reordering	Linear

Contrast - Angio

Flow Direction	F >> H
TONE Ramp	60 %

Resolution - Common

FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
Base Resolution	272
Phase Resolution	100 %
Slice Resolution	64 %
Trajectory	Cartesian
Interpolation	1.50

Resolution - Acceleration

Acceleration Mode	CS
Total Factor	12.0
Reference Scans	Integrated
Reference Lines PE	24
Reference Lines 3D	24
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Weak
Elliptical Scanning	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
POCS	Off
Distortion Correction	2D
Normalize	Prescan
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	8
Distance Factor	-22 %
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
Slices per Slab	64
Phase Oversampling	0 %
Slice Oversampling	12.5 %
FOV Read	200 mm
FOV Phase	91.2 %
Slice Thickness	0.30 mm
TR	20.31 ms

Geometry - Common

Multi-Slice Mode	Sequential
Series	Descending
Concatenations	8

Geometry - AutoAlign

Slab Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	R >> L
AutoAlign	Head > Brain
Initial Position	Isocenter
L	0.0 mm
A	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	90.00 deg

Geometry - Navigator**Geometry - Saturation**

Special Saturation	Tracking H
Gap	10.00 mm
Thickness	40.00 mm

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Performance
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg

System - Adjust Volume

A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	TONE

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.31 ms
Segments	1
Concatenations	8

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	200 mm
FOV Phase	91.2 %
Phase Resolution	100 %
Cine	Off
Trajectory	Cartesian
Dummy Heartbeats	1

Physio - PACE

Resp. Control	Off
Concatenations	8

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - Cardiac

Inline Evaluation	Off
Magn. Preparation	None
Save Original Images	On
Contrasts	1
TE	4.30 ms
TR	20.31 ms

Inline - MIP

MIP Sag	On
MIP Cor	On

Inline - MIP

MIP Tra	On
MIP Time	Off
Radial MIP	On
Number of Radial Views	12
Axis of Radial Views	H-F
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing

Inline Composing	Off
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Sequence - Part 1

Sequence Name	fl_r
Dimension	3D
Sequence Type	Gre
Excitation	TONE
RF Pulse Type	Normal
Gradient Mode	Fast
Flow Compensation	Slice/Read
Reordering	Linear
Bandwidth	188 Hz/Px
Echo Spacing	9.66 ms
Asymmetric Echo	Weak
Optimization	None
Define	Segments
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Phase Enc. Rewinder	On

Sequence - Assistant

SAR Assistant	TR
Max. TR	21.0 ms
Allowed Delay	0 s
Optimization	None